Artificial Intelligence Lab

Lab Report # 02

Submitted to:

Sir Mubashir Iqbal

Submitted By:

Shaheer Ahmad 23-cys-009

Dated:

March 2nd, 2026

Department of Cyber Security

HITEC University, Taxila

Python Lists, Tuples, Set and Dictionaries

Objectives: -

1. To understand basics of Python Lists, Tuples and Dictionaries.
2. To understand Python Functions
3. Implementing Python codes for lists, tuples, dictionaries and Functions.

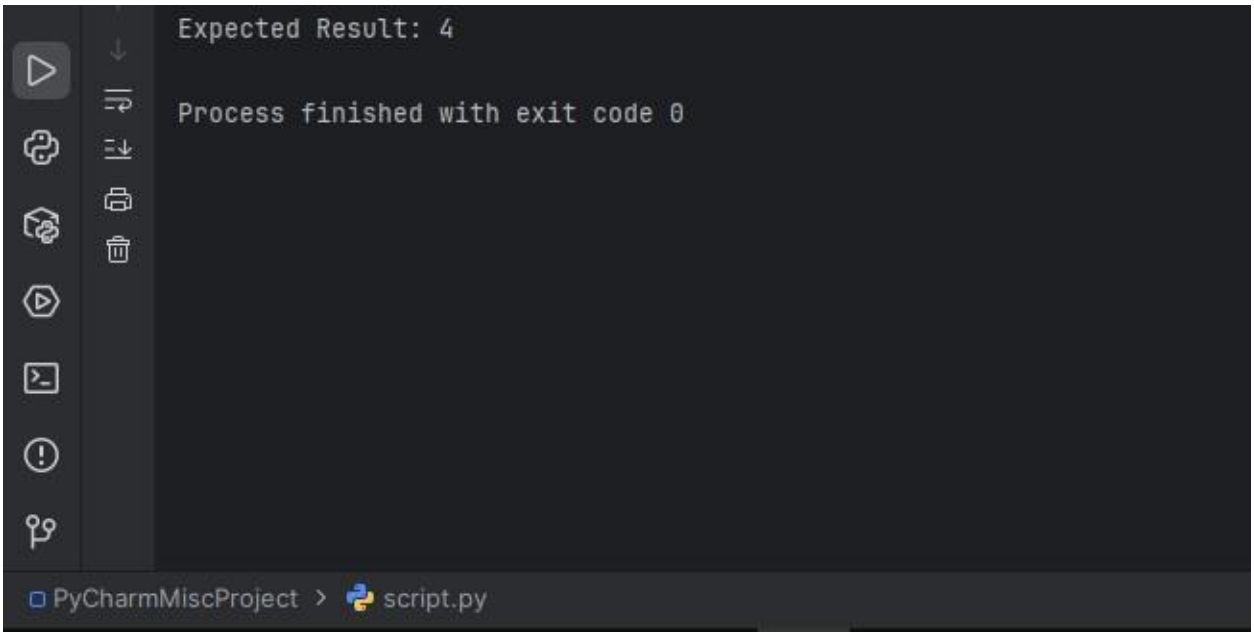
Lab Tasks

Question # 01

Count the number of strings where the string length is 2 or more and the firstand last character are same from a given list of strings. Sample List: ['abc', 'xyz', 'aba', '1221', 'xyzzyx', 'aa', '122’] Expected Result: 4

Solution:

Output: -


Expected Result: 4
Process finished with exit code 0
PyCharmMiscProject
script.py

Question # 02

Find the sum and product of numbers in a tuple. Sample List: (1, 2, 4, 2, 6), Sum is: 15, Product is: 96

Solution: -

```python
numbers = (1, 2, 4, 2, 6)

sum_numbers = sum(numbers)

product = 1
for n in numbers:
    product *= n

print("Sum is:", sum_numbers)
print("Product is:", product)
```

Output: -

```
Sum is: 15
Product is: 96

Process finished with exit code 0
```

Question # 03

Write a program to print largest number in a List Sample List: (1, 2, 4, 2, 6)

Solution: -

Output: -

Question # 04

Write a Python program to remove an empty tuple(s) from a list of tuples. Sample L = [(), (), ('',), ('a', 'b'), (), ('a', 'b', 'c'), (), ('d')], Result = [('',), ('a', 'b'), ('a', 'b', 'c'), ('d')]

Solution: -

Output: -

Question # 05

Write a Python script to generate and print a dictionary that contains a number (between 1 and n) in the form (‘x’, x*x). Sample: Input a number 4 Output: {'1': 1, '2': 4, '3': 9, '4': 16}

Solution: -

PyCharmMiscProject Version control

script.py

```python
n = int(input("Enter a number: "))
d = {}

for x in range(1, n + 1):
    d[str(x)] = x * x

print(d)
```

Output: -

```
Enter a number: 4
{'1': 1, '2': 4, '3': 9, '4': 16}

Process finished with exit code 0
```

PyCharmMiscProject > script.py

Question # 06

Create a program using Dictionary that takes Numbers as input and print it in words form Sample: Input: 1234, Output: One Two Three Four

Solution: -

```python
num = input("Enter a number: ")

words = {
    '0': 'Zero', '1': 'One', '2': 'Two', '3': 'Three', '4': 'Four',
    '5': 'Five', '6': 'Six', '7': 'Seven', '8': 'Eight', '9': 'Nine'
}

for digit in num:
    print(words[digit], end=' ')
```

Output: -

```
Enter a number: 43
Four Three
Process finished with exit code 0
```